

The Pennsylvania State University
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**CLUSTER ANALYSIS AND VISUALIZATION OF ROADWAY CONGESTION IN
CALIFORNIA**

Analyzing Spatial Autocorrelation of Congestion Cluster Snapshots in Six Major Cities

A Capstone in
Spatial Data Science

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ABSTRACT

This research examines the extent to which roadway congestion exhibits spatial clustering in six major California metropolitan regions using archived snapshots of real-time traffic data. Segment-level congestion metrics derived from the HERE Traffic API and OpenStreetMap road networks were analyzed using spatial statistical methods to quantify spatial dependence and identify localized clusters. Results demonstrate statistically significant, positive spatial autocorrelation for study areas, showing that congestion is spatially structured. To support interpretation, this research utilizes geovisual analytics (GVA) through the CongestionView CA application, which enables interactive exploration of spatial statistical outputs alongside underlying traffic conditions. By combining spatial statistics with visualization, the framework provides insights into congestion dynamics and decision making for transportation planning.

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1 INTRODUCTION

Traffic congestion is one of California's most enduring urban challenges, impacting public health, safety, and sustainability throughout the state. Congested road corridors contribute to lost time, increased emissions, and reduced quality of life for millions of residents. Populations and travel demand continue to grow, placing pressure upon transportation agencies to locate where congestion is most severe, how it changes through space and time, and which roadway types are most susceptible during peak travel periods.

Alongside evolving vehicular patterns, the sheer availability of traffic data has dramatically improved through online platforms, application programming interfaces (APIs), and open geographic datasets. This abundance of data offers new opportunities for monitoring and analysis, where raw traffic metrics are translated into actionable insights for transportation improvements; yet the effort of understanding these datasets remains a difficult process. Traditionally, diagnosing congestion on roadways requires the use of cumbersome traffic models, large-scale infrastructure, and counts taken in the field. As a result, there is a growing need for analytical approaches that can utilize existing data streams while reducing technical and financial barriers to entry.

This research places itself at the forefront of these issues by exploring the question of whether lightweight AI, spatial statistics strategies, and geovisual analytics (GVA) can be employed as low-cost, scalable tools for evaluating congestion metrics. Emphasizing the spatial structure of where congestion exhibits clustering behavior across metropolitan regions over predictive modelling or single location performance metrics provides an assessment for the variation in traffic patterns by roadway class and time of day. While this study focuses specifically on metropolitan regions within California, the methodological framework it develops is intentionally designed to be transferable to other locales. In doing so, the research aims to demonstrate how complex traffic datasets can be translated into accessible, interactive analytical insights that empower exploratory interpretation and decision-making, establishing a foundation for similar open and adaptable machine learning and GVA-based congestion analysis applications in other regions.

1.1 BACKGROUND

Recent research has established the potential of data-driven methods for analyzing roadway congestion without reliance on traditional traffic models or existing sensor infrastructure. Advances in cloud computing, the abundance of open data, and web-based traffic services have enabled researchers and traffic professionals to monitor congestion patterns using accessible data sources such as volunteered geographic information and online traffic analytics. Openly available traffic data and AI-powered analytics have been used to help lower barriers to entry for congestion monitoring in cities lacking traffic sensors or immense fiscal budgets, making analyses obtainable to planners with limited resources (Mirzahosseini et al., 2025). Beyond data sourcing considerations, effective traffic management tools must support temporal and spatial pattern recognition and predictive capabilities to inform urban traffic managers (Zhou & Hirasawa 2019). These studies suggest that traffic congestion displays structured spatial and temporal behavior that can be understood through analytics instead of uncorrelated or randomly occurring events stemming from localized, unconnected roadways.

A recurring theme is the importance of spatial components when utilizing heterogeneous data sources to explain complex urban events. Geographic context can serve as a unifying framework for linking unrelated datasets, a principle that is especially relevant for congestion studies integrating roadway geometries, classifications, and performance metrics (Hart & Dolbear 2007). Modern traffic management increasingly depends on incorporating multiple data streams to build spatially explicit traffic models, many of which can be sourced from openly available datasets such as OpenStreetMap

(Wu et al., 2025). Roadway congestion therefore appears to be not only time dependent, but also spatially influenced by network structure, road hierarchy, local travel demand, and other factors. While relevant, most of this work focuses on signal optimization and reducing carbon emissions; however, understanding the spatial structure is critical because it reveals how traffic relationships are shaped by underlying roadway networks and other dependencies. These components can only be effectively captured through the integration of diverse geospatial data sources such as road class, network structure, and real-time traffic metrics for road types and time of day.

Despite advances in methodology, prior research recognizes the difficulties in translating congestion trends derived from complex analyses into a digestible format for general audiences (Dodge et al., 2021). The application of advanced cartographic techniques for spatiotemporal visualization often requires specialized programming expertise and familiarity with data structures, which limits adoption by planners and non-technical stakeholders (Dodge et al., 2021). While deep learning-based traffic analysis pipelines have shown strong performance using scalable big data pipelines, such approaches depend on robust computational resources and expert-driven interpretation. This creates barriers to accessibility and transparency, particularly for non-technical stakeholders, and creates the need for lightweight, transparent, and easily explainable alternatives (Showribabu et al., 2025). Additionally, traffic studies often rely on single geographies, short-term and temporary traffic data, or individual congestion metrics, which presents a challenge when undertaking comparative analyses to quantify spatial dependence in multiple cities. While the granular scope of these studies is helpful for localized infrastructure improvements, measuring traffic distributions on a regional scale by combining spatial statistics, open data, and interactive visualizations better communicates where congestion clusters occur and how they vary by roadway class and travel periods. Aggregating congestion analyses for multiple metropolitan areas allows planners to identify consistent spatial patterns and cross-city trends, as the integration of diverse traffic datasets has been shown to increase the understanding of urban mobility patterns and support data driven transportation decision making (Mirzahosseini et al., 2025).

Traffic patterns vary dynamically, and exploratory visual analysis is beneficial for interpreting clustered behavior that may not be readily apparent from static charts or dashboards (Luo et al., 2023). Prior research indicates that identifying changes in traffic behavior, such as the emergence of congestion hotspots, is challenging using static analysis methods, and requires exploratory approaches supported by dynamic and historically rich datasets (van Landesberger et al., 2016). Because congestion hotspots emerge and dissipate cross roadway classes and peak travel periods, analysts require interactive tools that support exploratory comparison and investigation. Geovisual analytics (GVA) provides a methodological backdrop against which coordinated spatial and temporal views can be implemented, allowing users to move between regional and localized detail. Interactive spatial maps and modern visualization techniques enable exploratory analysis of traffic congestion, which allows users to transition from overviews to detailed inspection and facilitates a deeper understanding of complex patterns (Martinez & Bein, 2025).

Understanding where, when, and how traffic congestion occurs is important for designing transportation infrastructure in rapidly growing metropolitan regions. This research seeks to examine the spatial clustering of roadway congestion in six major California urban areas (Fresno, Los Angeles, Sacramento, San Diego, San Francisco, and San Jose) using archived snapshots of real-time traffic data from the HERE Traffic API combined with OpenStreetMap road networks. This research explores the value of lightweight and explainable analytic techniques grounded in spatial statistics and geovisual analysis. Integrating spatial autocorrelation measurements with interactive, coordinated visual interfaces enables the evaluation of whether traffic congestion exhibits measurable spatial dependence across California metropolitan regions and how clustering characteristics vary by roadway class and peak travel periods. As such, this research not only examines the spatial nature of congestion but also assesses the value of a

GVA-based interactive system for improving interpretability and decision-making in transportation planning.

1.2 RESEARCH QUESTION

This research develops an application called CongestionView CA to address current gaps by incorporating clustering methods that aim to provide interpretable, regional insights into the spatial structure of traffic congestion. The interactive design of CongestionView CA aids user navigation of complicated spatiotemporal clustering outputs through coordinated interactive views, helping contextualize statistical results within the underlying roadway network. Utilizing both spatial statistical measurements and interactive visual interfaces, this research advances the use of GVA as an analytical instrument to examine measurable spatial clustering in urban traffic systems.

To what extent does traffic congestion cluster spatially across road classes and peak travel periods in California metropolitan regions, and how effectively can Geovisual Analytics and machine learning methods detect and interpret these patterns?

2 METHODOLOGIES

2.1 STUDY AREA

The study area for this research focuses on six major metropolitan regions within California: Fresno, Los Angeles, Sacramento, San Diego, San Francisco, and San Jose (Figure 1). These cities represent the Central Valley, Southern California, and the San Francisco Bay Area, providing geographic, demographic and infrastructural diversity within a single statewide context. California was chosen due to the complexity and variability of its transportation systems and road network. The state contains a complex environmental landscape with coastal corridors, valleys, mountainous terrain, and deserts, each of which imposes differing constraints on traffic flow. Major roadway corridors such as I-5, US-101, and I-80 connect urban centers over long distances, generating interregional commuter and freight travel. Combining these factors together results in a spatially distinct picture of congestion patterns that makes California well suited for spatial data analysis.

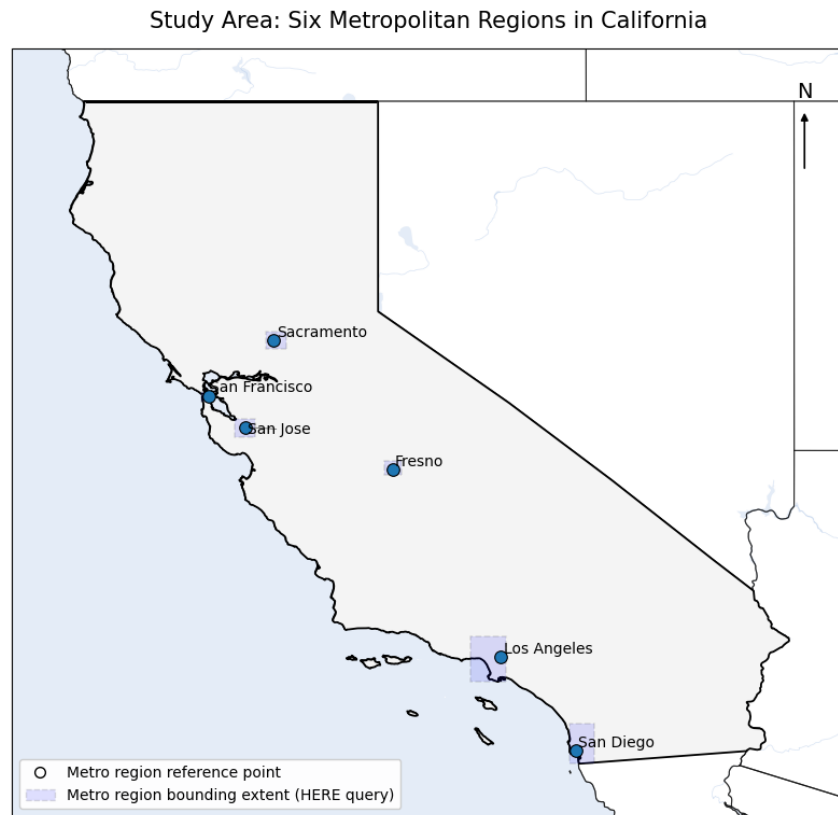


Figure 1: Study Area Map for Six Metropolitan Regions (EPSG:4326)

2.2 DATA SOURCES

The two primary datasets used to capture urban mobility patterns in the six Californian urban areas were the OpenStreetMap (OSM) roadway network data and HERE Traffic Flow Application Programming Interface (API) data.

OpenStreetMap (OSM) is a free, crowd-sourced map database that represents a global road infrastructure network updated by volunteers online (Haklay & Weber, 2008). OSM provides openly accessible vector data representing roads, buildings, and other geographic features. For this research, roadway geometries were extracted using the *OSMnx* Python library, which downloads and converts OSM data into a GeoDataFrame format that can be manipulated programmatically (*OSMnx Documentation*). OSM data is created and maintained by a global community of contributors who digitize and update features using aerial imagery, local knowledge, and municipal data sources. Edits are subject to ongoing peer review within the OSM ecosystem, especially in highly populated areas where mapping activity is frequent. The six metropolitan areas studied in this research are extensively mapped and frequently updated due to their population size and general local activity levels.

While OSM is a continuously updated dataset with street data and other features going back to 2004, the current basemap represents infrastructure as it is currently seen by contributors (*History of OpenStreetMap*, 2025). Road network data was downloaded during the 2025 study period for this research to coincide with the analyzed traffic flow snapshots. OSM represents relatively static infrastructure components, so temporal data matching concerns are minimal compared to constantly changing traffic data. For this research, only drivable, paved road classifications were retained within the predefined metropolitan extents, while footways, bike lanes, and service alleys were excluded to ensure relevance for vehicular congestion monitoring. Key OSM attributes utilized include roadway

name and highway classification; road class is used to evaluate congestion variability across infrastructure types.

The HERE Traffic API, an online service offered by HERE Technologies, provides real-time traffic flow and incident data in JSON format. HERE Technologies aggregates traffic information from a combination of connected vehicle probes, GPS-enabled fleet data, mobile device pings, roadway sensors, and other proprietary sources (HERE Technologies, 2023). These data sources are processed into traffic flow metrics available via the API. The free-tier license offers a maximum of 5,000 monthly transactions; when the API is called iteratively through each urban study region, this limit is sufficient given the temporal sampling period under investigation, which consists of multiple daily traffic snapshots for the project area over a two-week period (HERE Technologies, 2023). While the HERE API places several usage restrictions in its agreement for the use of its base plan, the intended use as an analytic tool for this research does not violate these guidelines. The platform does not track individual assets in real time, nor does it send route guidance, and usage falls into a data analytics context which is an accepted use of the HERE Traffic API data under the terms of service (HERE Technologies, 2023).

For this research, real-time traffic flow conditions were captured during defined AM and PM peak commuting periods for a two-week period in 2025 to evaluate temporal variability in congestion behavior. Every traffic snapshot includes timestamp metadata recording the moment of data retrieval and congestion conditions at that time. To ensure data matching, all metropolitan regions were queried during the same temporal timeframes for each collection occurrence. This approach ensures that regional comparisons reflect consistent time of day traffic conditions unilaterally. HERE traffic data is returned as vector polyline geometries representing roadway segments with traffic attributes, which are matched to OSM-derived road geometries and enriched through nearest spatial joins. HERE Technologies provides data coverage internationally and for the entire continental United States; for this research, traffic data is requested using bounding boxes corresponding to each metropolitan study area, only capturing segments falling within those defined extents (HERE Technologies, 2023).

Since the unit of analysis is the roadway segment, extracted from OSM and enriched with traffic attribution from the HERE API, several key datapoints from each source are central to this research:

- **Free Flow Speed:** The reference speed (in meters per second) along the roadway when no traffic is present. This indicates how fast traffic would move under ideal conditions, such as late at night with no cars.
- **Speed:** The expected speed (in meters per second) along the roadway. This represents how fast vehicles are currently moving, or predicted to move, based on live traffic data.
- **Jam Factor:** The amount of traffic on a roadway, with values ranging from 0.0 to 10.0 indicating quality of travel. HERE does not publicly disclose a calculation formula for this metric, however it is theorized to be influenced by many factors such as speed vs. free flow baseline, traffic volume, incident reports, and local sensor data. Values are grouped in the following congestion levels: 0 - 4: No or light traffic; 4 - 8: Moderate; 8 - 10: Heavy traffic; 10: Blocked roadway.
- **Speed Ratio:** Calculated as Speed divided by Free Flow Speed, representing relative travel efficiency.
- **Road Classification:** OSM defined highway type used to analyze congestion patterns for different infrastructure types.
- **Segment Centroids:** Spatial coordinates generated from roadway segment geometries to enable spatial statistic testing.
- **Temporal Snapshots:** Timestamps associated with traffic data when exported during AM and PM peak commuting time periods to examine temporal variability in congestion patterns.

2.3 DATA TRANSFORMATION

After defining the six key metropolitan regions for study, each urban area was queried using a defined bounding box expressed in WGS84 geographic coordinates. These bounding boxes were manually specified to encompass each metropolitan core and primary commuter corridors while avoiding inclusion of extraneous rural areas. OSM roadway segment geometries were reprojected to EPSG:3857 (meters) for accurate distance calculations, and invalid geometries, such as line segments with null or empty attributes from spatial joins, were then removed to ensure graph connectivity and accurate spatial weighting. This data cleaning process ensures that each road segment is captured with valid spatial coordinates and that all features were connected to the neighborhood graph. Because the derived variables extracted from the HERE Traffic API operate on different numerical scales, a standardization using z-scores was applied to prevent scale dominance during the clustering procedure.

To support spatial testing and clustering, a composite congestion index was derived from the HERE Traffic API variables. Exploratory assessment of the Jam Factor and Speed Ratio indicators was conducted to understand their distributional characteristics and levels of association. Since Jam Factor reflects an absolute congestion intensity index, while Speed Ratio captures efficiency loss, both variables showcase overlapping measurements of congestion conditions. To create an indicator for unified spatial statistical testing in this research, a composite congestion index is calculated as:

$$(1 - \text{Speed Ratio}) + \text{Jam Factor}$$

This derived metric combines relative speed reduction and congestion intensity into a single index, making it quantitatively grounded and useful for spatial autocorrelation analysis. The composite congestion score is constructed to support the clustering algorithm and spatial statistical methods, as it represents a synthesis of the raw congestion metrics instead of an independently observed variable. Higher composite congestion scores imply more severe congestion conditions, like increased jam conditions and reduced travel efficiency, while lower values correspond to free-flowing segments. Because the metric is derived from standardized traffic measurements, comparisons can be made throughout regions and temporal snapshots.

2.4 EXPLORATORY DATA ANALYSIS

Exploratory spatial data analysis (ESDA) was performed on the raw congestion indicators, Jam Factor and Speed Ratio, to evaluate the distributional characteristics and spatial structure prior to their integration into the composite congestion index. Exploratory analysis revealed that congestion levels in the urban Californian project areas are low to moderate on average during the single day collection period (Sunday at 2:00PM ET) but exhibit a strong relationship between key congestion indicators. Descriptive statistics confirmed low to moderate congestion conditions universally, with consistent inverse correlation between Jam Factor and Speed Ratio (r ranging from -0.82 to -0.92), validating that these two variables not only measure related aspects of congestion but are also complementary components of traffic conditions without redundancy (Table 1).

Table 1: Summary Statistics for Jam Factor Indicator by City

Region	Number of Segments	Mean	Median	Standard Dev	Skew	Correlation Coefficient (Speed Ratio)
Fresno	1251	1.009	0.8	1.036	1.234	-0.925
Los Angeles	12109	1.155	0.9	1.238	1.643	-0.850
Sacramento	2502	0.988	0.7	1.171	1.861	-0.848

San Diego	5280	1.027	0.5	1.355	2.309	-0.825
San Francisco	2388	1.412	1.2	1.366	1.469	-0.838
San Jose	3677	0.958	0.6	1.118	1.573	-0.870

Global Moran's I statistics, which measure the extent to which similar values occur near one another more frequently than under spatial randomness, demonstrated statistically positive spatial autocorrelation when considering both jam factor and speed ratio for all areas (I from 0.12 to 0.24, $p < 0.001$) (Tables 2 & 3). This process confirmed that, while the scale of autocorrelation was moderate, congestion is not randomly distributed but spatially clustered along roadway corridors. As such, the consistency of positive spatial dependence through all regions, as evidenced by the uniformly positive and spatially significant Moran's I values, supports the application of clustering and spatial statistics in this research.

Table 2: Global Moran's I for Jam Factor Indicator by City

Region	Number of Segments	Moran's I (Jam Factor)	P Value (Jam Factor)	Z Score (Jam Factor)
Fresno	1251	0.208	0.001	15.563
Los Angeles	12109	0.185	0.001	42.331
Sacramento	2502	0.173	0.001	19.329
San Diego	5280	0.243	0.001	37.277
San Francisco	2388	0.120	0.001	12.604
San Jose	3677	0.190	0.001	25.621

Table 3: Global Moran's I for Speed Ratio Indicator by City

Region	Number of Segments	Moran's I (Speed Ratio)	P Value (Speed Ratio)	Z Score (Speed Ratio)
Fresno	1251	0.214	0.001	16.061
Los Angeles	12109	0.229	0.001	54.336
Sacramento	2502	0.225	0.001	25.603
San Diego	5280	0.236	0.001	36.077
San Francisco	2388	0.144	0.001	14.547
San Jose	3677	0.193	0.001	25.949

2.5 CONGESTION CLUSTERING METHODS

Unsupervised K-Means clustering was employed to classify roadway segments into three congestion groupings representing relatively low, moderate, and high congestion conditions within each temporal snapshot. K-Means is an algorithm that assigns observations to a predefined number of clusters by minimizing variance within clusters, grouping segments with similar traffic conditions into classes. This approach was selected because congestion is not defined by a single metric, but the interaction of multiple variables. In the context of this research, K-Means clustering serves as an important step that transforms continuous traffic measurements into categorical configurations which can then be evaluated for spatial structure. Clustering was applied to the combined dataset of roadway segments across all six

regions, rather than being trained separately for each city. By pooling observations into a single clustering model, the algorithm identifies congestion groupings based on patterns present for the entire study extent, ensuring consistency in how clusters are defined.

Features incorporated in the model included Jam Factor, Speed Ratio, spatial coordinates, and road classification, allowing examination of congestion by both infrastructure and location contexts. To better reflect the importance of congestion intensity and travel efficiency, feature weighting was applied: jam factor was multiplied by 5.0 and speed ratio by 10.0, emphasizing their influence in defining congestion, while roadway classification was assigned a lower weight of 1.0 to preserve its spatial influence without controlling cluster formation. These weighted variables were used as inputs to the clustering algorithm following feature standardization, which was applied to prevent dominance by variables with larger numeric ranges. A three-cluster configuration was chosen to represent low, moderate, and high congestion levels. The number of clusters (k) was not fixed at the outset, and multiple k values were tested and evaluated based on cluster coherence and interpretability. To enable direct comparison across time, the final configuration maintained a consistent number of three clusters. Cluster labeling was performed dynamically based on the cluster's mean composite congestion index, ensuring grouping labels reflect relative congestion conditions within each temporal snapshot.

Cluster outputs represent relative congestion levels within each metropolitan region and timeframe. A "Congested" label signifies that segments in that cluster exhibit higher jam intensity and reduced speed efficiency relative to other segments in the same snapshot. Because the clustering process is unsupervised, group boundaries are driven by metrics rather than absolute definitions; as such, clusters should be interpreted as comparative congestion measurements rather than fixed categories. The clustering methodology reflects previous research in that effective traffic management systems should provide recognition of spatial and temporal patterns, anticipate emerging congestion tendencies, and provide status assessments for urban decision makers (Zhou & Hirasawa, 2019). By using aggregated snapshots of real-time metrics and standard transportation attributes, the model outputs explainable congestion groupings that align with these objectives.

2.6 SPATIAL STATISTICAL APPLICATION

Following the K-Means clustering technique, spatial clustering analysis of the congestion behavior was performed to evaluate whether the observed congestion characteristics exhibited measurable spatial dependence. Spatial autocorrelation techniques were selected to determine if similar congestion conditions occur in proximity more frequently than under random spatial distribution, addressing the objective of identifying clustering within roadway networks.

Global spatial autocorrelation was assessed using Moran's I , which provides a single statistic describing the overall degree of dependence between each metropolitan region. Moran's I measures the correlation between congestion values at a given location and those of its neighboring segments, detecting systematic clusters at the network level. Spatial relationships were defined through a k -nearest neighbors (KNN) weights matrix ($k=8$) constructed from the roadway centroids. Neighbors were assigned based on Euclidean proximity rather than network topology; as a result, neighboring segments identified in the weights matrix did not necessarily share direct physical connections with the roadway network but instead represented close spatial segments within the urban infrastructure. This process ensured that there were a consistent number of neighbors for each observation and avoided issues related to network fragmentation; however, it is possible that relationships may be introduced between segments that are geographically close but not connected, such as parallel roads or grade-separated travelways. Considering this, the KNN approach was chosen to capture broader spatial clustering patterns on road corridors rather than strictly modelling network flow between segments.

The composite congestion score was used as the central variable for all spatial autocorrelation tests. Similarly to the input features, row standardization was applied to the spatial weights matrix to ensure comparability, and statistical significance for Moran's I was evaluated using permutation testing (9,999 permutations). Moran's I values range from -1 to +1, where positive values indicate clustering of similar congestion levels (high near high, low near low) and negative values represent spatial distribution (high near low). Statistical significance was determined at a threshold of $p < 0.05$, however, given the large number of roadway segments (high n), statistically significant results may occur even when the magnitude of Moran's I is relatively low. Therefore, both the coefficient magnitude and significance level must be considered equally when assessing spatial dependence.

To complement the global analysis, Local Moran's I statistics were determined for each roadway segment to identify specific locations where congestion was significantly clustered or spatially dissimilar. While Moran's I evaluates overall spatial dependence, LISA breaks this global measurement down into localized relationships. By detecting local clusters of congested roadways that are near other congested roadways, this process allows for an understanding of spatial characteristics within each metropolitan area. Each roadway segment is evaluated relative to the average composite congestion score of its defined neighborhood, with significance determined via permutation testing at a threshold of $p < 0.05$. However, because Local Indicators of Spatial Association involve performing many simultaneous statistical tests, this method is inherently more susceptible to Type I error, where some segments may appear significant by chance alone. While more conservative corrections can be applied to control for multiple comparisons, the $p < 0.05$ threshold was enforced in this study to balance statistical rigor with interpretability, as stricter corrections may overly suppress meaningful spatial patterns in large datasets. Therefore, LISA results should be interpreted as indicative of spatial clustering tendencies instead of definitive statistical boundaries. This consideration further supports the utilization of geovisual analytics, where statistical outputs are interpreted alongside spatial context.

The Local Moran's I analysis output evaluates whether each feature's value is similar or different from the average of its neighbors: positive values indicate similar congestion clusters, while negative values indicate local congestion variances. The combined use of global and local spatial autocorrelation techniques results in a broad assessment of clustering behavior and a more detail examination of localized spatial patterns, creating a more comprehensive understanding of congestion for the regional analysis.

LISA outputs classify segments into one of five categories:

- High-High: Highly congested segment located near other highly congested segments
- Low-Low: Low congested segment located near other low congested segments
- High-Low: Congested segment located near uncongested segments
- Low-High: Uncongested segment located near congested segments
- Not Significant: No significant spatial relationship

Clusters categorized as High-High indicate that corridors experience spatially concentrated congestion and therefore serve as priority zones for further evaluation by stakeholders. High-Low and Low-High segments reveal localized inconsistencies in network performance.

2.7 TEMPORAL CONGESTION COMPARISON

For this research, traffic flow data was captured during defined morning and evening peak hour commuting windows to assess temporal variability in congestion levels. Over a 13-day window from March 17 to March 29, 2026, snapshots from the HERE and OSM data sources were collected and stored at 7 AM, 8 AM, and 9 AM PT, as well as 4 PM, 5 PM, and 6 PM PT. These time intervals were

selected to represent typical peak travel periods commonly associated with commuter activity; additionally, data collection was conducted continuously for both weekdays and weekends. As a result, the temporal analysis reflects a combination of weekday commuting patterns and weekend travel behavior. While weekday traffic patterns may be more strongly associated with traditional commuting flows, weekend observations provide additional context for understanding how congestion structures persist or vary under different travel demand conditions.

The resulting traffic data, which contained real-time values for road segments at the time of capture, was subsequently processed through the clustering pipeline. By applying identical clustering and spatial statistical methods to each snapshot, changes in congestion grouping and spatial dependence patterns could be compared over time. Temporal variation was assessed by comparing changes in cluster size, shifts in Moran's I values, and variation of High-High or Low-Low LISA clusters. An increase in Moran's I during peak periods would represent stronger corridor-level congestion concentrations, while unchanging values across time would indicate persistent traffic bottlenecks.

2.8 GEOVISUAL ANALYTICS FRAMEWORK

Traffic congestion exhibits variation by time and space, and statistical outputs alone may not fully convey emergent patterns or shifts over time (Luo et al., 2023). Geovisual analytics (GVA) provides a foundation for adding quantitative spatial analysis to interactive visual interfaces. In this research, spatial clustering outputs, such as the K-Means classifications, composite congestion scores, and LISA outputs are embedded in an interactive mapping environment called CongestionView CA. The CongestionView CA dashboard was developed using CARTO Builder, a cloud-based geospatial analytics platform that enables interactive mapping, data integration, and spatial analysis workflows (CARTO, 2025). Linking spatial statistical elements with dynamic cartography, the platform provides an evaluation of spatial dependence, roadway classification, and temporal variability for users. The benefit of lightweight and explainable analytics in this research is demonstratable by showing that spatial structure can be identified without relying on expensive or computationally heavy software and instead can promote transparency and reproducibility in transportation decision-making.

The GVA system for this research includes coordinated visual components to support exploratory analysis of congestion clusters and spatial statistics, with outputs symbolized by color and line width in a mapping application to represent cluster severity. Summary graphics act as a complement to the cartographic elements, with real-time counts of roadways segments by congestion cluster, histograms illustrating Speed Ratio distribution values, time series visualizations showing shifts in cluster membership throughout the project period, and attribute inspection through segment specific pop-up windows. The GVA interface does not replace statistical testing but serves as an interpretive window through which statistical output can be contextualized for non-technical users. Through maps and diagrams accessible on the CongestionView CA portal, significant clustering relationships can be visually assessed for practical use in transportation planning.

3 RESULTS

The following sections present the results of the clustering and statistical analyses applied to the collected traffic snapshots, which were taken three times during the morning commute (7AM, 8AM, 9AM PT) and evening commute (4PM, 5PM, 6PM PT) over the thirteen-day collection period. The cumulative analysis focuses on patterns of spatial autocorrelation, evaluating variation over metropolitan regions, and examining differences between peak travel times.

3.1 CONGESTION CLUSTERING

The clustering model produced roadway segment classifications of Free Flowing, Moderate Flow, and Congested, allowing the distribution of congestion classes to be summarized for the six metropolitan regions. These results describe how the modeled congestion classes were distributed within each study area over the entire study period prior to the spatial statistical analysis. It is important to note that the clustering summary statistics presented in this section are aggregated for all temporal snapshots collected during the study period. Specifically, congestion classifications were generated for six time intervals per day over a 13-day collection window, resulting in a total of 78 snapshots per metropolitan region. As such, individual road segments may be represented multiple times in the summary, corresponding to their classification at different time intervals. Overall, the congestion class summaries varied considerably in both total number and composition of classified roadway segments throughout each city.

The distribution of congestion classes differed systematically across all metropolitan regions (Figure 2 & Table 4). Fresno, Sacramento, San Francisco, and San Jose were all dominated by the Moderate Flow classification, indicating that most roadway segments in these regions experienced intermediate congestion conditions throughout the cumulative study period. This indicates relatively balanced traffic conditions, where segments were neither consistently free flowing nor constantly congested. Conversely, Los Angeles and San Diego exhibited a fundamentally different pattern, with Free Flowing segments comprising the largest share of classifications. This distinction signifies that, despite their larger roadway networks and higher overall traffic volumes, these regions contained a greater proportion of segments operating under relatively uncongested conditions at any time. The divergence between these two groups of cities highlights regional differences in congestion conditions: some metropolitan areas are characterized by widespread moderate congestion, and others display a more distinct distribution between free flowing and congested states. Because Figure 2 and Table 4 represent raw counts, these differences also reflect variation in the number of roadway segments included within each study area. Los Angeles was associated with the largest segment inventory due to the broader metropolitan bounding box used for data collection and the greater density of vehicular roadways captured within that extent.

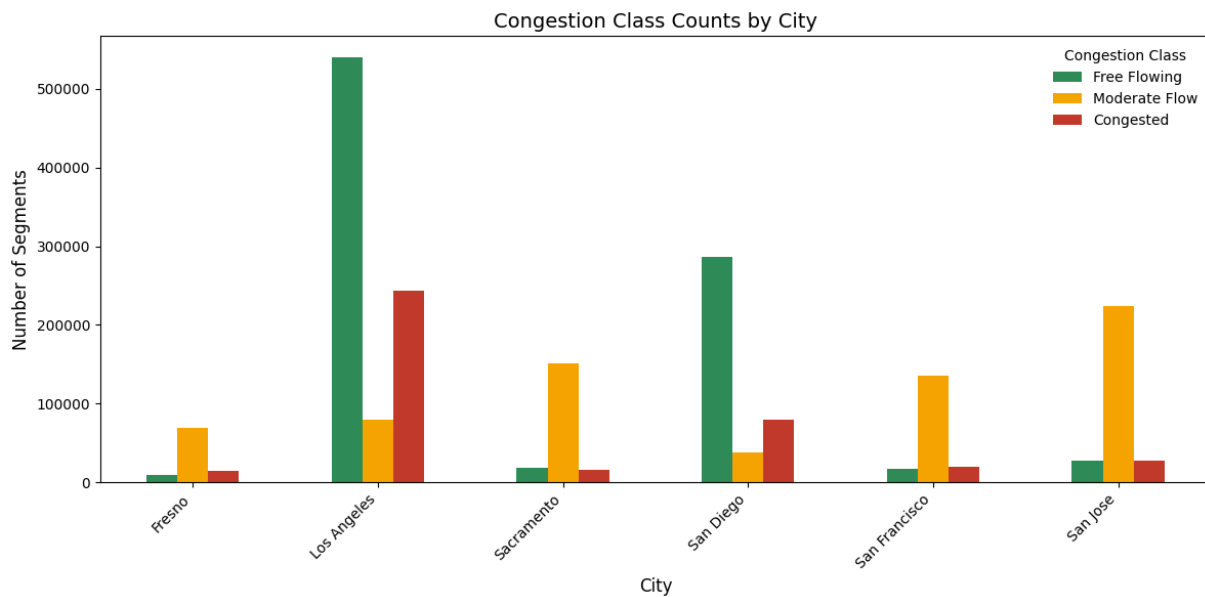


Figure 2: Congestion Class Breakdown by City

Table 4: Summary Statistics for Congestion Clustering by City

City	Free Flowing	Moderate Flow	Congested	Total	Free Flowing (%)	Moderate Flow (%)	Congested (%)
Fresno	9697	69750	14598	94045	10.3	74.2	15.5
Los Angeles	539722	79222	243100	862044	62.6	9.2	28.2
Sacramento	18411	151493	15837	185741	9.9	81.6	8.5
San Diego	286812	37671	79392	403875	71	9.3	19.7
San Francisco	17139	136192	20139	173470	9.9	78.5	11.6
San Jose	27268	223387	27717	278372	9.8	80.2	10

When visualized, proportional distributions further highlight variation by city and travel period. For this analysis, AM and PM classifications were aggregated for all days in the study period, with AM representing the combined results from three morning snapshots, and PM representing three evening snapshots collected daily over the data period. This aggregation provides a generalized representation of peak-period congestion behavior, reducing short term variability and emphasizing persistent classification patterns.

In Fresno, Sacramento, San Francisco, and San Jose, congestion classifications were regularly led by the Moderate Flow category in both AM and PM periods (Figure 3 & Table 5). This pattern shows that roadway conditions in these urban areas tend to remain within an intermediate congestion range, as opposed to shifting towards extremes of free flowing or congested states during peak periods. Notably, this distribution remains stable between the morning and evening travel times, suggesting that congestion in these regions is broadly dispersed across the network, and is not concentrated within specific time windows. In contrast, Los Angeles and San Diego displayed congestion patterns with substantially larger shares of Free Flowing classifications in both AM and PM periods, distinguishing them from the moderate-controlled distributions observed in the other cities. This pattern implies that a larger proportion of roadway segments in these cities operate under relatively uncongested conditions across all peak travel periods, despite their larger roadway networks.

Overall, the AM and PM distribution results in relatively limited variation of congestion classification patterns between peak travel periods in these cities. Because temporal differences between morning and evening conditions are less pronounced than underlying spatial structure, congestion appears to persist at similar levels throughout the day rather than exhibiting strong directional shifts associated with commuter flows.

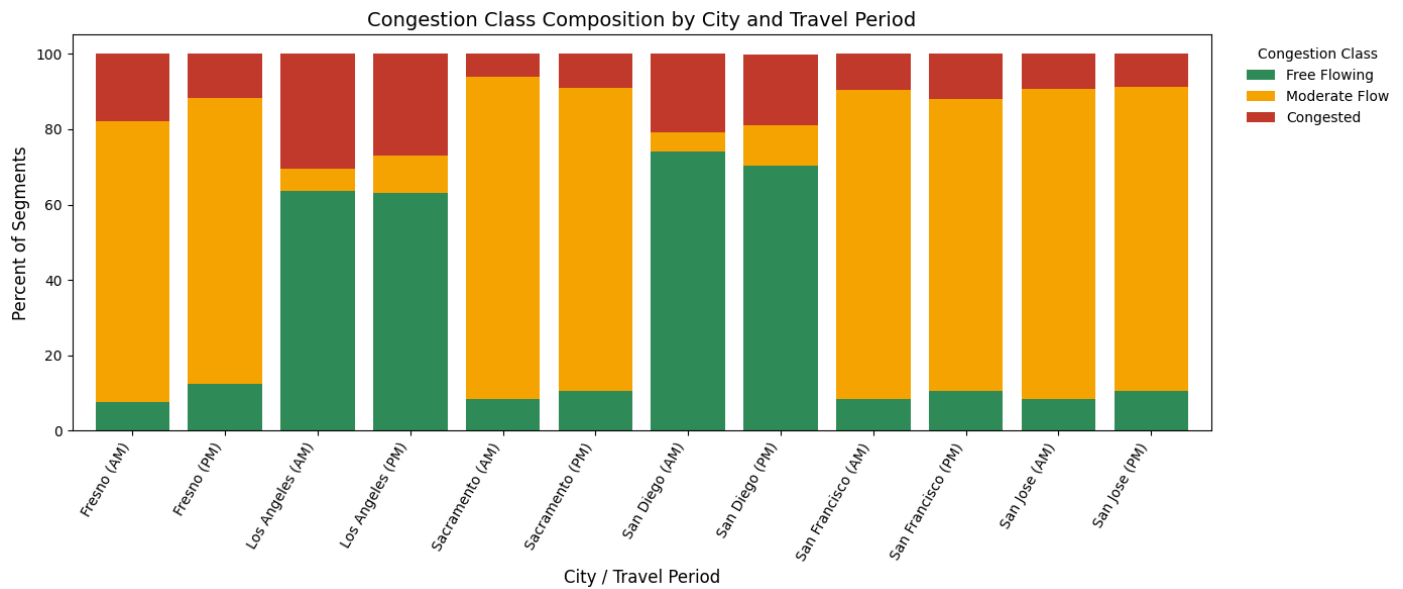


Figure 3: Congestion Class Breakdown by City and Travel Period

Table 5: Summary Statistics for Congestion Clustering by City and Time Period

City	Travel Period	Free Flowing	Moderate Flow	Congested	Total	Free Flowing (%)	Moderate Flow (%)	Congested (%)
Fresno	AM	3160	31467	7627	42254	7.5	74.5	18.1
Fresno	PM	5341	33000	5007	43348	12.3	76.1	11.6
Los Angeles	AM	245930	23369	117680	386979	63.6	6	30.4
Los Angeles	PM	251108	39735	106878	397721	63.1	10	26.9
Sacramento	AM	7007	71145	5197	83349	8.4	85.4	6.2
Sacramento	PM	9080	68834	7821	85735	10.6	80.3	9.1
San Diego	AM	134061	9347	37695	181103	74	5.2	20.8
San Diego	PM	131199	19988	35343	186530	70.3	10.7	18.9
San Francisco	AM	6535	63790	7399	77724	8.4	82.1	9.5
San Francisco	PM	8463	62004	9740	80207	10.6	77.3	12.1
San Jose	AM	10438	102874	11665	124977	8.4	82.3	9.3
San Jose	PM	13468	103805	11171	128444	10.5	80.8	8.7

3.2 SPATIAL STATISTICS

Global spatial autocorrelation was evaluated using Moran's I to quantify the overall degree of clustering within each metropolitan region. When aggregated for all temporal snapshots, Global Moran's I values ranged from approximately 0.10 to 0.16 (Table 6). This aggregation represents 78 total snapshots per urban area, derived from six daily time intervals collected over the project period. These observations include both weekday and weekend conditions, meaning that the reported statistics represent a combined

assessment of peak-period traffic behavior in varying travel situations. As such, the aggregated Moran's I values can be interpreted as generalized indicators of spatial clustering for peak travel periods, as opposed to strictly filtering for weekday only commuter-driven congestion patterns.

For all metropolitan regions and time intervals, Moran's I values were consistently statistically significant ($p < 0.05$), with most observations exhibiting strong significance levels ($p \approx 0.001$ or lower). This uniform significance implies that spatial autocorrelation was a demonstrable characteristic of the dataset, rather than being driven by isolated or non-significant observations. Fresno and San Diego experienced the highest average spatial autocorrelation, while San Francisco and San Jose displayed comparatively lower values and more spatially diffuse congestion structures. All cities exemplified consistent positive spatial dependence, reinforcing the presence of non-random congestion patterns in diverse urban environments. Additionally, the similarity between mean and median Moran's values confirms that spatial clustering patterns were stable throughout the study period and were not caused by extreme outlier observations.

Table 6: Global Moran's I values aggregated by city during analysis period

City	Mean Moran's I	Median Moran's I	Mean # Segments	# Snapshots
Fresno	0.167	0.170	1,206	78
Los Angeles	0.134	0.138	11,052	78
Sacramento	0.119	0.108	2,381	78
San Diego	0.151	0.144	5,178	78
San Francisco	0.116	0.116	2,224	78
San Jose	0.105	0.097	3,569	78

Disaggregation by travel period revealed minor differences in spatial autocorrelation between morning (AM) and evening (PM) peak periods (Table 7). In all metropolitan regions, Moran's I values were higher during PM periods compared to AM periods, showing stronger spatial clustering of congestion during the evening commute. In contrast, slightly lower Moran's I value during AM periods present a more dispersed distribution of congestion, potentially demonstrating more mixed travel patterns, including arrival trips and non-work-related travel. The magnitude of AM to PM differences varied by region. Fresno, Sacramento, and San Francisco exhibited the most pronounced temporal contrasts, possibly due to stronger corridor-driven commuting tendencies. In comparison, San Diego and San Jose showed smaller differences between time periods and more distributed or network-wide congestion behavior. San Francisco presented comparatively lower Moran's I values, particularly during the AM period (mean I = 0.093), signifying relatively weak spatial autocorrelation despite statistical significance. This suggests that congestion patterns in this context are more diffuse and did not form clearly defined spatial clusters.

Table 7: Global Moran's I values aggregated by city and travel time during analysis period

City	Travel Period	Mean Moran's I	Median Moran's I	Mean # Segments	# Snapshots
Fresno	AM	0.14	0.143	1207	39
Fresno	PM	0.194	0.197	1205	39
Los Angeles	AM	0.124	0.127	11052	39
Los Angeles	PM	0.144	0.152	11051	39

Sacramento	AM	0.106	0.099	2381	39
Sacramento	PM	0.131	0.131	2382	39
San Diego	AM	0.144	0.144	5175	39
San Diego	PM	0.158	0.149	5181	39
San Francisco	AM	0.093	0.096	2220	39
San Francisco	PM	0.139	0.14	2228	39
San Jose	AM	0.103	0.104	3570	39
San Jose	PM	0.107	0.093	3568	39

Positive autocorrelation in congestion for the metropolitan study regions persisted in both travel periods and was reinforced by the boxplot figures for AM and PM dispersal (Figure 4). These figures show meaningful differences in variation, with Fresno and San Diego exhibiting elevated medians and wider ranges in the PM peak periods, resulting in more fluctuating congestion conditions. Comparatively, Los Angeles has a generally stable distribution and is moderately clustered with a smaller interquartile range despite having a much larger roadway network. San Jose consistently demonstrates lower median values, particularly in the evening, and weaker overall clustering relative to the other cities. Several outliers are present in Los Angeles, Sacramento, San Francisco, San Diego, San Francisco, and San Jose, where some individual snapshots diverged considerably from the typical level of spatial dependence. These observations endorse that congestion clustering was consistently present in all cities, and its magnitude and stability changes by location and travel period.

The several low-value outliers present in the metropolitan regions reveal instances where spatial autocorrelation was notably weaker than the typical range. One potential explanation for these observations is the inclusion of weekend data within the temporal sample. Weekend travel patterns are generally less structured than weekday commuting flows by nature, often characterized by more dispersed trip origins or varied travel purposes, which may result in less pronounced spatial clustering during these periods. Because the analysis aggregates both weekday and weekend observations, these lower Moran's I values may be a symptom of underlying differences in travel behavior, and not random variation. While this study does not explicitly separate weekday and weekend patterns, future analysis could isolate these temporal subsets to further examine how travel behavior influences the strength of spatial autocorrelation.



Figure 4: Distribution of Global Moran's I values by city for all temporal snapshots

To further examine spatial dependence at the city level, Moran's I scatterplots were analyzed for individual snapshots. The Fresno Moran scatterplots for 4 PM, 5 PM, and 6 PM provide a clear example of persistent positive spatial autocorrelation during the PM peak period (Figures 5 – 7). For all three snapshots the regression line maintains an upward slope with Moran's I values within a narrow range (0.197 – 0.245) as roadway segments with relatively high congestion tend to be located near similarly congested neighbors. This resulted in a right-skewed distribution, where a small number of higher-value observations contributed to the overall positive spatial autocorrelation.

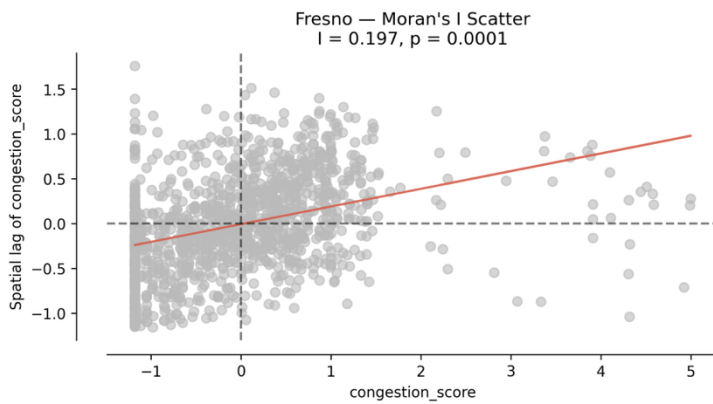


Figure 5: Global Moran's I scatterplots for clustered Fresno road segments (3/26/26), 4 PM

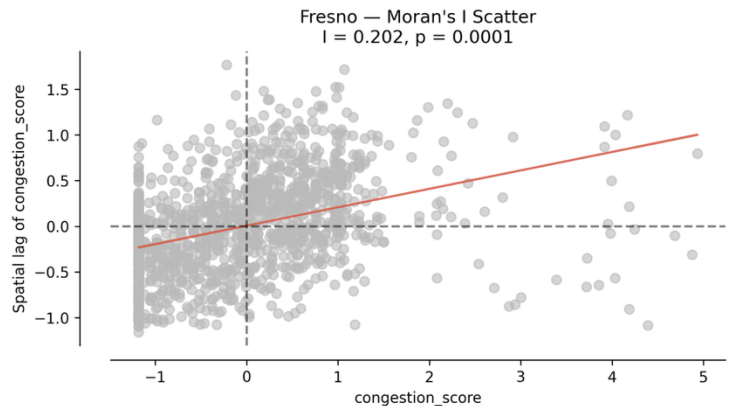


Figure 6: Global Moran's I scatterplots for clustered Fresno road segments (3/26/26), 5 PM

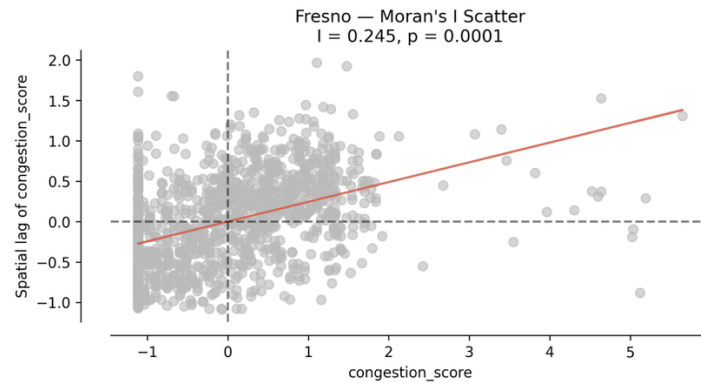


Figure 7: Global Moran's I scatterplots for clustered Fresno road segments (3/26/26), 6 PM

Comparing a single time slice for multiple metropolitan regions allows for a regional comparison of congestion clustering at different magnitudes; as all study regions are statistically significant, a gradient from stronger to weaker positive spatial autocorrelation is present. At 9:00 AM on Monday, March 23, Fresno, Los Angeles, San Diego, and San Francisco exhibited differing strength of cluster variation by region (Figures 8 - 11). Fresno showed the strongest positive relationship ($I=0.194$), followed by San Diego ($I = 0.172$), while Los Angeles ($I=0.135$) and San Francisco ($I=0.116$) displayed weaker but significant clustering. The scatterplots display a positive slope consistent with statistically significant spatial autocorrelation, yet the degree of dispersion around the regression line varies by city. In several cases, the relatively high degree of scatter suggests weaker spatial dependence, which is reflected in the lower Moran's I values observed for some cities. This signifies that, although spatial clustering is present, the strength of the relationship between neighboring segments is not uniformly strong in all metropolitan areas.

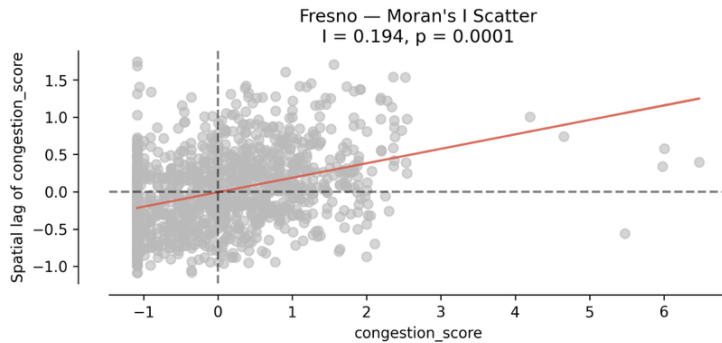


Figure 8: Global Moran's I scatterplots (3/23/26, 9 AM) for Fresno

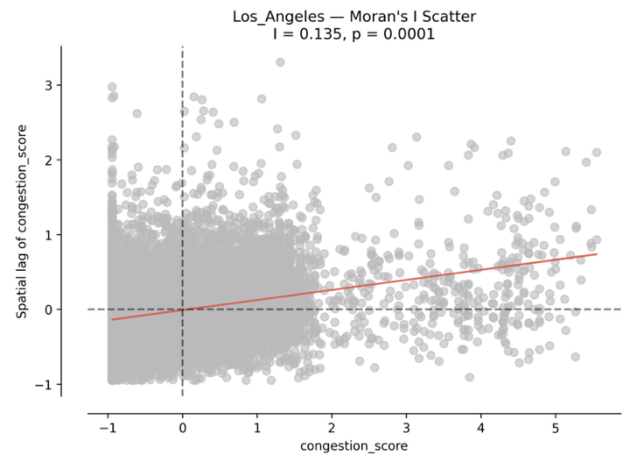


Figure 9: Global Moran's I scatterplots (3/23/26, 9 AM) for Los Angeles

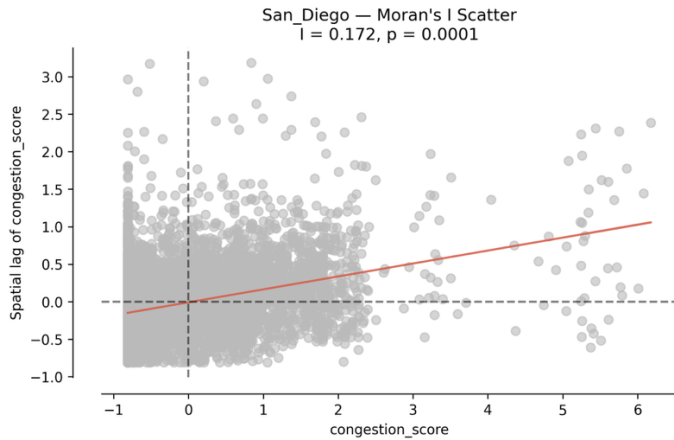


Figure 10: Global Moran's I scatterplots (3/23/26, 9 AM) for San Diego

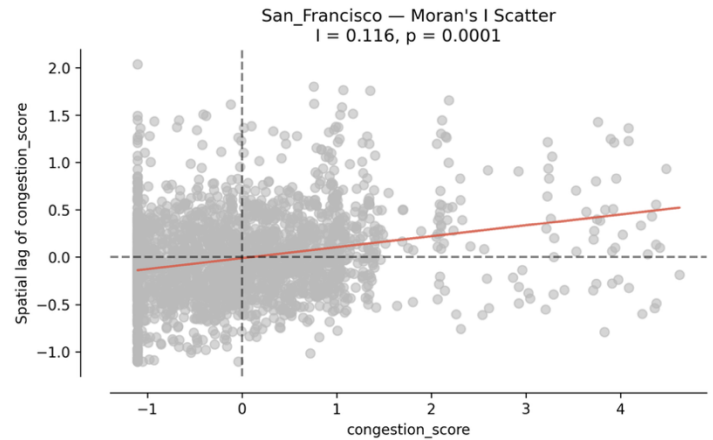


Figure 11: Global Moran's I scatterplots (3/23/26, 9 AM) for San Francisco

Local Moran's I (LISA) analysis was used to identify the spatial distribution of statistically significant congestion clusters within each metropolitan region. The resulting maps revealed consistent patterns of localized spatial dependence, providing an understanding into where congestion is concentrated and how it propagates throughout the roadway network. The Los Angeles evening peak period offers a representative example of these dynamics, particularly when comparing the 5:00 PM and 6:00 PM snapshots on Thursday, March 19th (Figures 12 & 13). In both maps, High-High (HH) clusters, or segments with high congestion surrounded by similarly congested neighbors, appeared as localized concentrations within dense portions of the roadway network. These HH patterns remained visible between the 5:00 PM and 6:00 PM snapshots, suggesting that some congestion relationships persisted during the evening period, where high congestion values were spatially reinforced by nearby roadway segments with similarly elevated congestion conditions.

Comparatively, Low-Low (LL) clusters reflected areas where lower congestion values were surrounded by similarly low congestion neighbors. These areas exhibited more stable free flow conditions within portions of the network. Transitional cluster types, including High-Low (HL) and Low-High (LH), were observed along the boundaries between differing congestion conditions. These classifications highlight areas of rapid spatial transition in congestion conditions, where localized inefficiencies or spillover effects may occur between highly congested and uncongested segments. Although the overall clustering structure remained relatively consistent between 5:00 and 6:00 PM, the redistribution of transitional clusters implies that localized congestion effects were dynamic within the broader pattern of evening congestion.

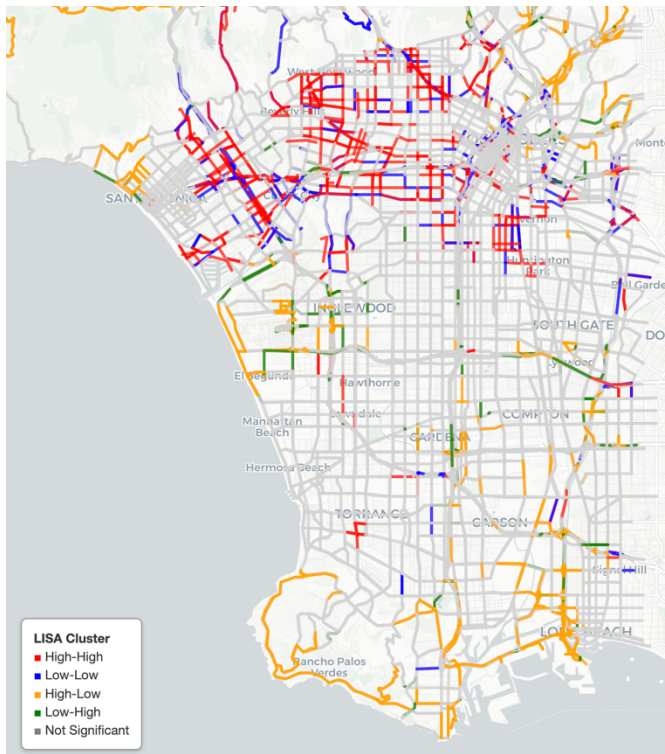


Figure 12: LISA cluster map for Los Angeles (3/19/26, 5 PM)

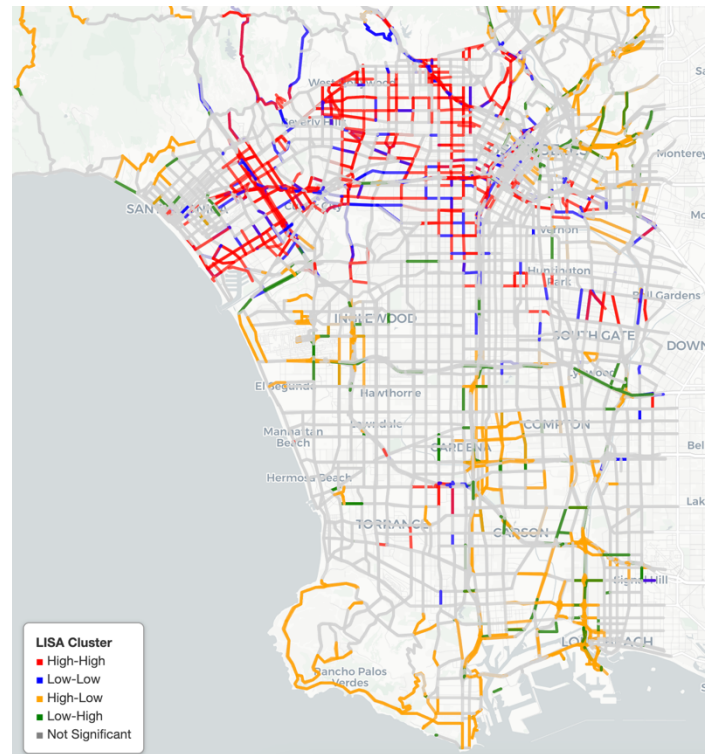


Figure 13: LISA cluster map for Los Angeles (3/19/26, 6 PM)

Temporal comparison of LISA outputs across multiple days revealed that congestion clustering in San Francisco was both recurring and spatially dynamic (Figures 14 & 15). Assessment of the 8:00 AM snapshots from Wednesday, March 18th and Thursday, March 19th showed several core cluster locations persisting, particularly within central urban corridors and segments approaching the eastern portion of the network, confirming the presence of recurring congestion hotspots. At the same time, the March 19 map displayed a broader and more pronounced set of significant clusters than March 18, signifying that the spatial footprint of congestion expanded under similar morning peak conditions.

The comparison also exposed clear shifts in cluster composition between the two mornings. While High-High (HH) clusters remained anchored in similar parts of the network, the March 19 snapshot had a greater spread of significant HH, Low-Low (LL), and transitional clusters at adjacent road segments. Recurring hotspots remained in place while surrounding segments experienced expansion or transition in local congestion relationships due to a redistribution of congestion within the network, as opposed to reflecting random fluctuation.

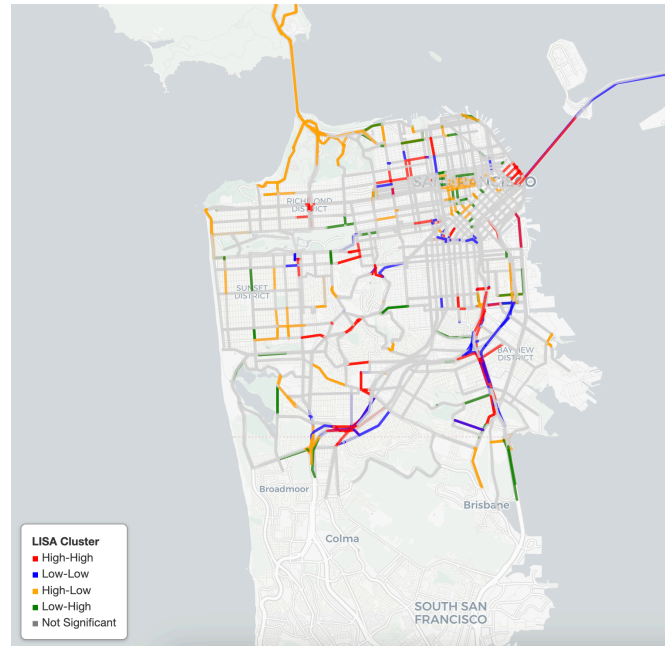
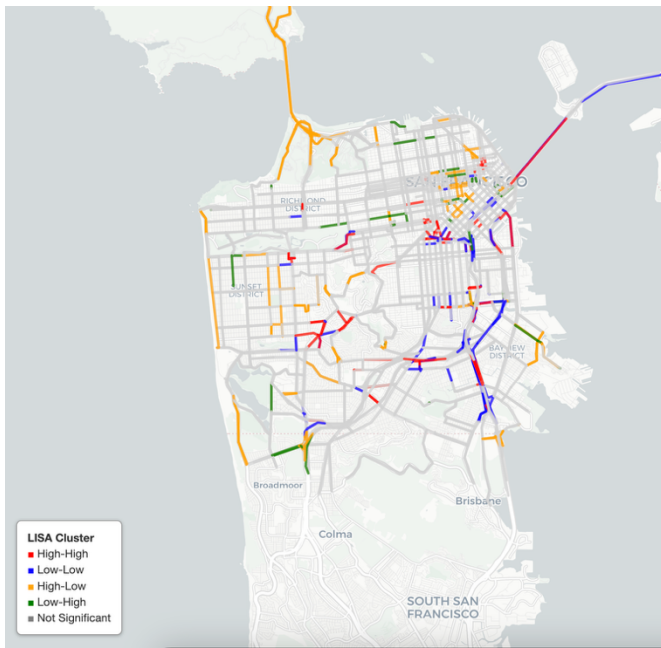


Figure 14: LISA cluster maps for San Francisco (3/18/26, 8 AM)

Figure 15: LISA cluster maps for San Francisco (3/19/26, 8 AM)

To further evaluate whether local spatial association patterns varied by roadway hierarchy, LISA cluster membership was summarized by OSM road class for the full set of temporal snapshots (Figure 16). This comparison adds additional perspective beyond the maps, which show the location of LISA clusters but not do not directly depict roadway classification. Throughout most road classes, non-significant segments represented the largest share of observations, signifying that statistically significant local spatial association was concentrated within a subset of the roadway network, rather than uniformly distributed across all segments. Among significant observations, High-Low relationships were frequently the dominant LISA type for city and road class combinations, while High-High clusters appeared more selectively in certain road types and regions. This result refines the interpretation of the LISA maps. Rather than showing that congestion hotspots are concentrated exclusively on major arterials or freeway classes, the road class comparison suggests that local congestion structure often takes the form of transitional spatial relationships. High-High clusters remain important for identifying localized groupings of severe congestion, but High-Low and Low-High categories reveal contrasts between adjacent roadway segments. These transitional zones may be especially meaningful for transportation planning because they identify locations where congestion may emerge or interact with nearby free flowing segments. As a result, roadway classification should be interpreted as contextual information for understanding LISA patterns, not a standalone explanation for cluster formation.

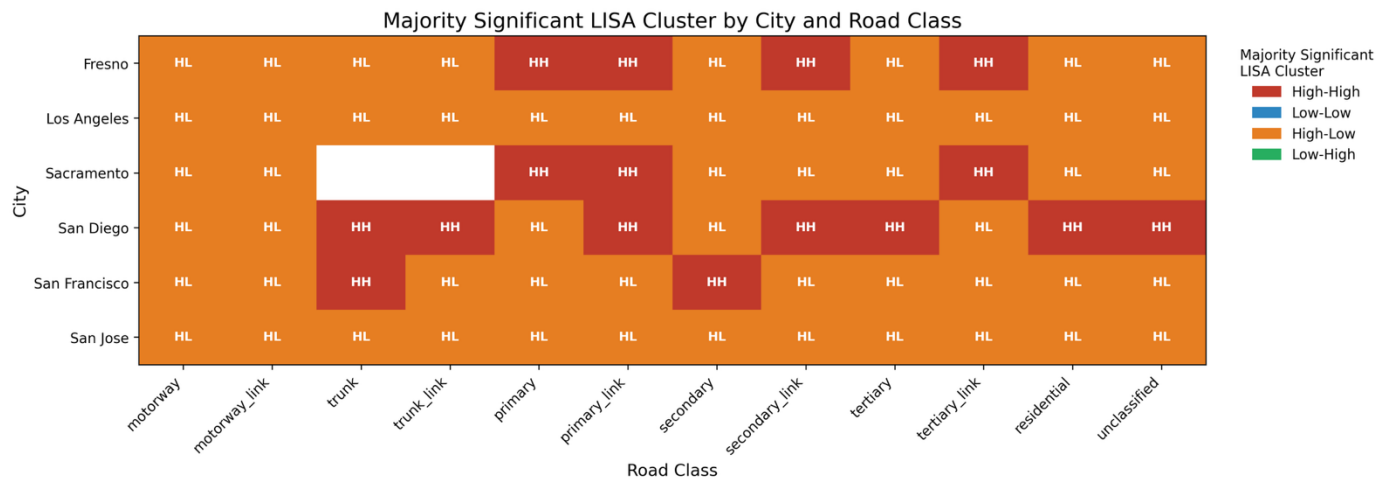


Figure 16: Majority significant LISA cluster type by city and OSM road class

Overall, Global Moran's I values confirmed that congestion maintained a spatial structure for all cities and snapshots, while LISA analysis revealed the localized hotspot, coldspot, and transitional relationships that shaped each metropolitan network. Temporal comparisons showed that these clustering patterns were dynamic rather than fixed, with the spatial extent and configuration of congestion shifting by travel period and day. Additional comparison of LISA cluster membership by roadway classification further explored these observations by showing that significant local spatial relationships were not confined to a single road class. Instead, High-High clusters appeared in certain city and roadway combinations, with High-Low relationships frequently the dominant significant LISA type for all road classifications. This indicates that local congestion structure was generally transitional, contrasting between congested and less congested neighboring segments. Collectively, these findings support the use of spatial statistical methods and clustering techniques as effective tools for identifying roadway congestion patterns, while also establishing a quantitative foundation for the CongestionView CA geovisual analytics framework.

3.3 GEOVISUAL ANALYTICS

Drawing upon the results of the spatial statistical analysis, an interactive geovisual analytics (GVA) system titled CongestionView CA was developed, facilitating user exploration of congestion patterns for the six urban study areas. The platform consists of two coordinated dashboards: a clustering dashboard and a spatial statistics dashboard, each designed to support complementary aspects of the analysis. The clustering dashboard features roadway segments classified using the K-Means clustering algorithm with dynamic filtering tools, temporal controls, and attribute-level inspection, presenting congestion clusters using a three-class symbology scheme (low, medium, high) to identify spatial concentrations of congestion throughout the road network (Figure 17). In parallel, the spatial statistics dashboard visualizes Local Indicators of Spatial Association (LISA) results, symbolizing segments by cluster type (High-High, Low-Low, etc.) and incorporating similar interactive components to explore localized spatial relationships (Figure 18). Additional interface elements across both dashboards, including filtering widgets and pop-up windows, support deeper inspection of individual segments and their associated traffic values.

The CongestionView CA dashboards serve as visual extensions of the clustering and spatial statistical outputs, providing interactive environments in which patterns can be explored in greater detail. Accessible in tandem, the clustering dashboard enables users to interpret relative congestion groupings of roadway segments, and the spatial statistics dashboard allows for direct exploration of localized

spatial relationships identified through LISA analysis. Together, these interfaces allow users to contextualize statistical results geographically and observe how clustering and spatial dependence manifest along specific corridors and intersections.

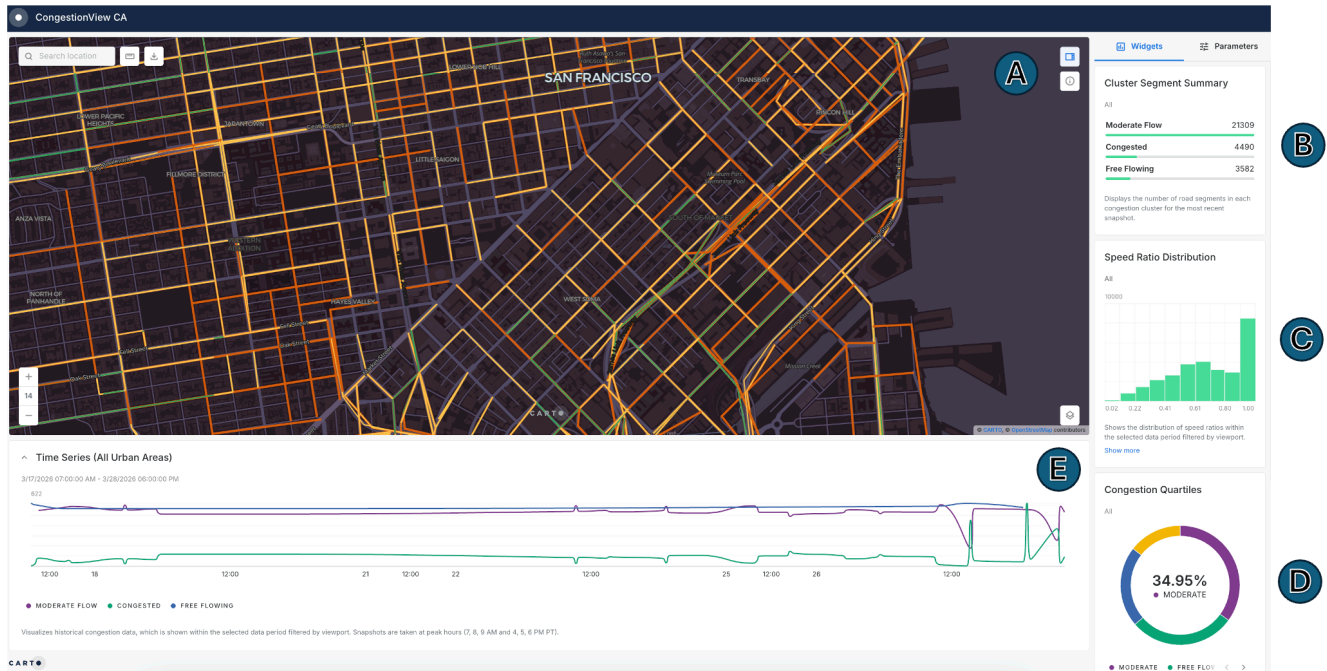


Figure 17: Main interface for the CongestionView CA clustering dashboard. (A) Interactive map displaying congestion clusters, (B) Cluster summary statistics widget, (C) Speed Ratio distribution histogram, (D) Congestion quartiles pie chart, (E) Time series visual

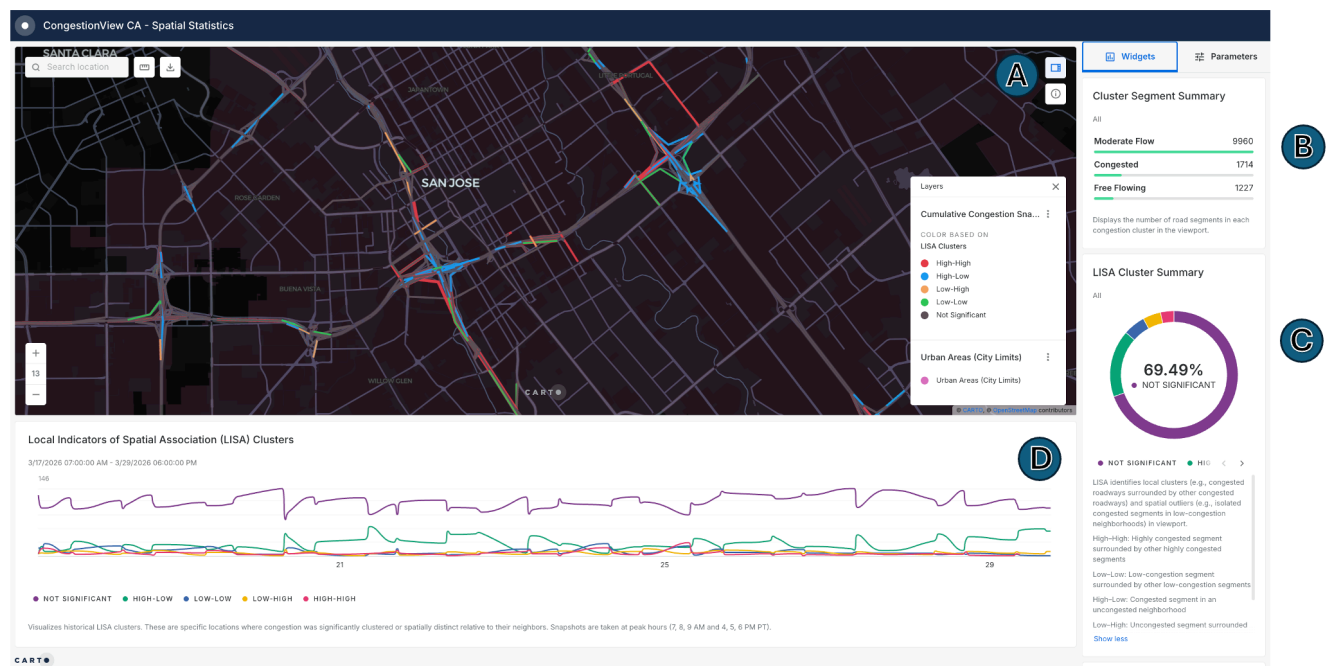


Figure 18: Main interface for the CongestionView CA spatial statistics dashboard. (A) Interactive map displaying LISA clusters, (B) Cluster summary statistics widget, (C) LISA clusters pie chart, (D) Time series visualization of LISA cluster membership

Both dashboards contain filtering capabilities that allow users to subset the data by metropolitan region, time period, and roadway classification, allowing for direct comparison of congestion patterns by spatial

and temporal dimensions. Applying filters to isolate individual cities, specific travel periods, and road types reveals distinct spatial distributions of congestion clusters and spatial statistical groupings that may be hidden in aggregated views. For example, high-congestion clusters within urban downtown areas in a city like Fresno appear more spatially extensive and interconnected, whereas LISA outputs reveal how these segments are embedded within broader patterns of spatial dependence (Figure 19 & Figure 20). All visual components within the dashboards are dynamically linked to the map interface through coordinated views, so that summary statistics, charts, and other visualizations update in response to both attribute-based filters and changes in the map extent. As users zoom or pan to different geographies, the displayed metrics automatically recalculate to show only the visible roadway segments, supporting contextual exploration of spatial relationships.

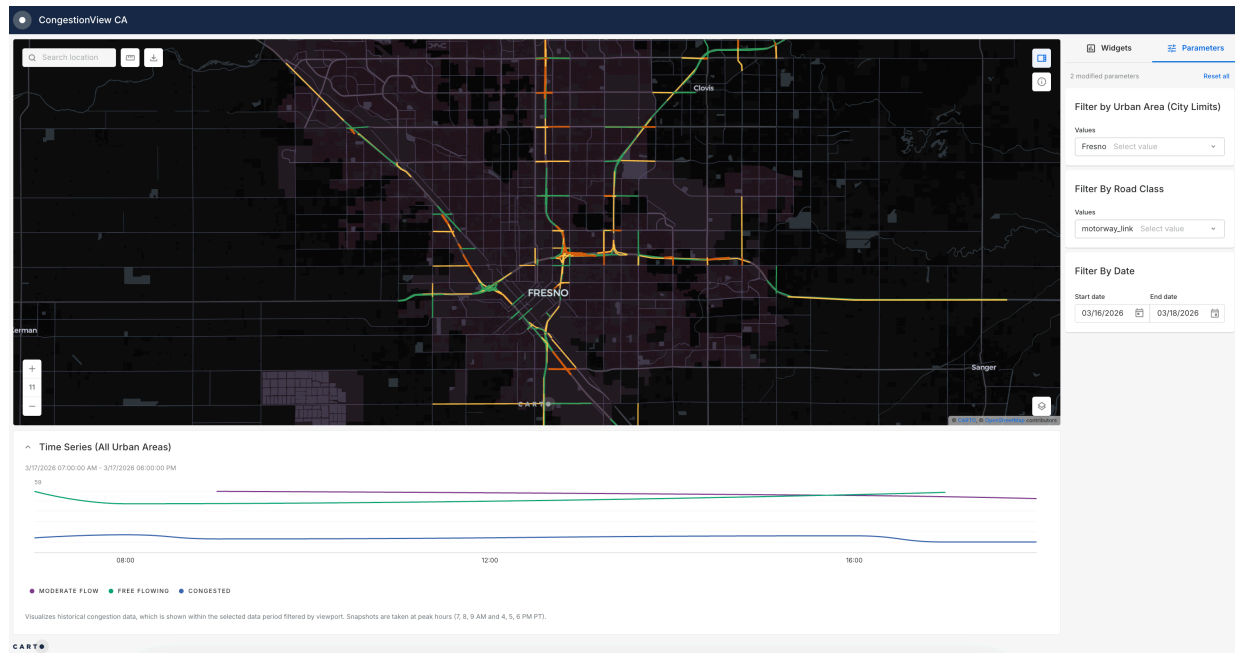


Figure 19: Filtered view by city (Fresno) targeting congestion clusters for motorway links captured for 3/17/26 in congestion dashboard

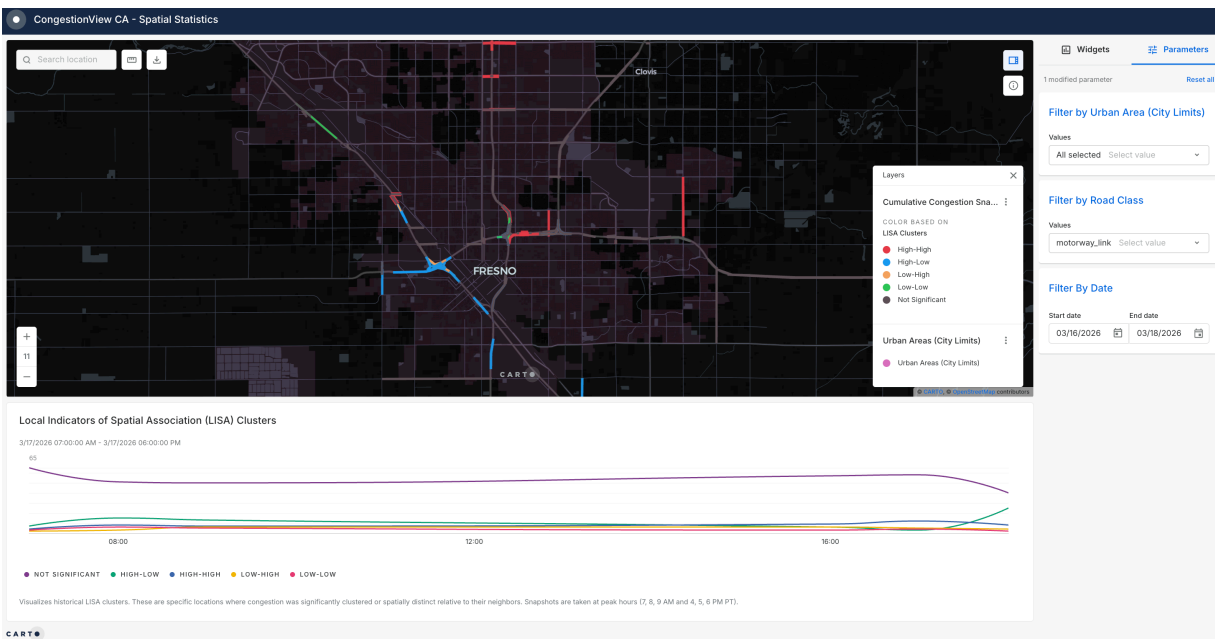


Figure 20: Filtered view by city (Fresno) targeting LISA clusters for motorway links captured for 3/17/26 in spatial statistics dashboard

In addition to spatial filtering, the dashboards include temporal controls and time-series visualization components that allow users to study how traffic patterns evolve over the study period. Temporal visualization summarizes changes in congestion intensity, clustering classifications, and spatial dependence throughout all selected snapshot intervals, providing insight into the consistency and variability of roadway corridors on screen (Figure 21). The clustering dashboard emphasizes changes in congestion grouping; at the same time, the spatial statistics dashboard highlights how the strength and structure of spatial relationships evolve over time.

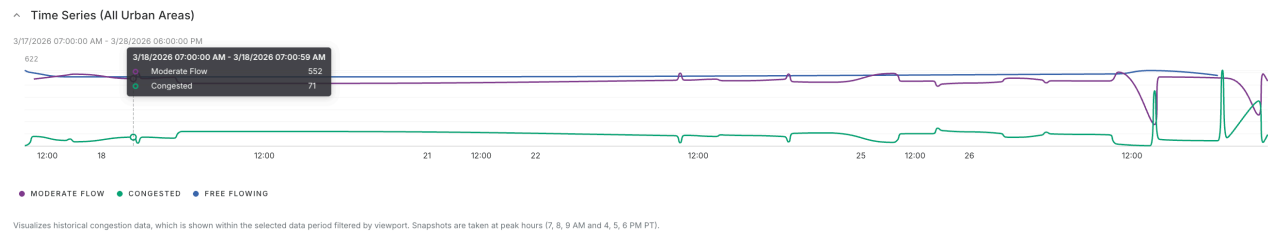


Figure 21: Time series and temporal filtering element included in congestion and spatial statistics CongestionView CA dashboard interfaces

While the spatial statistics analysis evaluates clustering and spatial dependence at certain time points, the temporal functionality of the GVA dashboards provides continuous user-driven engagement with congestion dynamics. This empowers users to identify recurring patterns, such as persistence of high-congestion clusters along major corridors during peak travel hours, as well as shifts in spatial relationships or anomalous observations over time. By integrating both clustering outputs and spatial statistical results within coordinated visual interfaces, the GVA framework extends analytical value by linking quantitative findings to broader spatial and temporal trends in an interactive and interpretable format.

4 DISCUSSION

This research establishes that roadway congestion in select California metropolitan regions exhibits consistent and measurable spatial clustering, supporting the contention that congestion is not randomly distributed but follows structured spatial patterns. In all six study areas, positive spatial autocorrelation was observed, and that highly congested roadway segments tend to be located near similarly congested neighbors. These patterns were further shown to vary by time of day, with evening peak periods producing stronger and more concentrated clustering than morning conditions, due to more corridor-driven congestion during PM commutes. While the magnitude and distribution of clustering differed by city, the emergence of these patterns across all regions highlights the influence of network structure and travel behavior on congestion dynamics. In addition to confirming the presence of spatial dependence, the combination of clustering methods, spatial statistics, and geovisual analytics proved effective in detecting and interpreting these patterns, proving the value of easy to use, explainable analytical approaches for understanding complex traffic systems.

The findings of this research provide a clear response to the research question by validating both the presence of spatial clustering in traffic congestion and the effectiveness of the analytical framework used to evaluate it. Spatial statistical results demonstrate that congestion exhibits positive spatial autocorrelation and that roadway performance is influenced by neighboring segments rather than occurring as isolated events. Local Indicators of Spatial Association (LISA) further reveal that congestion is concentrated in identifiable hotspot corridors, while transitional cluster types highlight areas of interaction between congested and uncongested segments and the interconnected nature of roadways. The clustering model provided an interpretable classification of roadway conditions, while the interactive visualization environment enabled users to contextualize outputs within their geographic setting and explore variation over time in congestion levels. These results determine that both components of the research question were addressed, and that congestion clustering can be quantitatively measured alongside the use of geovisual analytics for practical applications.

The observed spatial patterns imply that congestion is strongly affected by underlying travel behavior and roadway network structure as opposed to occurring as a uniform condition within metropolitan areas. Consistently higher levels of spatial autocorrelation during evening peak periods indicate that congestion becomes more spatially concentrated as travel demand congregates along primary commuting corridors. This pattern reflects the directional nature of end of day travel, where large volumes of vehicles are focused onto a limited set of arterial routes and highway segments. In contrast, comparatively lower and more diffuse clustering observed during morning periods suggests a broader distribution of trip origins and more variable travel purposes, resulting in less concentrated spatial congestion patterns. Differences between metropolitan regions further reinforce that congestion clustering was not identical within the study area. Fresno and San Diego generally exhibited stronger mean Moran's I values, suggesting more cohesive clustering of similar congestion values during the observation period. By contrast, San Francisco and San Jose recorded comparatively weaker spatial autocorrelation, with San Francisco's AM results indicating especially limited spatial structure. These differences indicate that congestion clustering may be shaped by how regional travel demand interacts with local roadway form; however, because this study did not directly measure network centrality or corridor dependence, these explanations should be interpreted as plausible contextual factors as opposed to confirmed causality.

Further insight into congestion properties can be found when considering roadway classification and local spatial relationships identified through LISA analysis. Rather than demonstrating that significant congestion clusters were confined to a single roadway type, the road class comparison showed that local spatial association differed over the roadway hierarchy. High-High clusters remained important for

identifying localized groupings of severe congestion, but they did not evenly dominate where traffic demand is expected to be highest, like major arterials and freeways. High-Low relationships were instead more frequently the majority significant LISA type in many city and road class combinations, indicating that congestion often appeared as a transitional spatial relationship between congested and less congested neighboring segments. Summarily, this substantiates the interpretation of the LISA maps by showing that congestion structure is not only defined by concentrated hotspots, but also by local contrasts within the roadway network.

Highlighting these transitional relationships may be especially meaningful from a planning perspective. Whereas High-High clusters point to areas where congestion is spatially reinforced, High-Low and Low-High clusters emphasize locations where congestion conditions change across nearby roadway segments. Such areas may correspond to ramps, intersections, merges, road class transitions, or other localized network features where traffic behavior shifts over short distances. Therefore, the LISA results provide a pretense for planners to examine both hotspot clusters and spatial transition zones when evaluating congestion, and the role of GVA in this research is strengthened because interactive dashboards allow users to inspect segment level attributes, road class, and spatial statistics together rather than relying on a single static map or metric.

The geovisual analytics framework developed in this research plays a critical role in translating the statistical outputs into interpretable and actionable insights. While measures such as Moran's I and LISA provide quantitative evidence of spatial dependence, their meaning is not always immediately accessible without spatial context. The CongestionView CA interface addresses this gap through two complementary CARTO dashboards: one focused on K-Means congestion cluster outputs and another focused on LISA-based spatial statistical outputs. Together, these dashboards embed analytical results within interactive environments that support exploration by location, time period, roadway classification, congestion cluster, and LISA cluster type. Coordinated filtering, temporal controls, map-linked widgets, and segment level pop-ups allow users to move between regional views and localized detail, making it possible to examine how congestion and local spatial relationships evolve within specific corridors or intersections. The GVA framework doesn't function as a formal outlier detection tool, yet it supports exploratory inspection of temporal changes and localized deviations that warrant closer review through the underlying statistical outputs. In doing so, the GVA applications enhance the interpretability of this research and promote exploratory reasoning, aligning with the "overview first, details on demand" principle and demonstrating how complex spatial patterns can be communicated effectively to non-technical audiences (Mason et al., 2025).

The dashboards demonstrate the technical feasibility of integrating clustering outputs, spatial statistics, and interactive visualization; however, the benefits of the GVA system were not formally evaluated through user testing. A full assessment of the system's effectiveness would require structured evaluation with intended users, such as transportation planners or GIS analysts, to determine whether the dashboards improve interpretation or support decision-making compared to traditional maps and tables. Future work could address this through usability studies in which participants complete defined analytical tasks like identifying recurring congestion hotspots, comparing AM and PM clustering patterns, or locating LISA cluster groupings. Additional evaluation could include user confidence scores and perceived usefulness through surveys or interviews. In its current state, the GVA component should be regarded as a proof-of-concept visualization framework, with formal usability and decision support evaluation representing an important area for growth.

This research contributes to the growing body of literature on data-driven traffic analysis by addressing different methods identified in prior studies. Previous work has highlighted the challenges of interpreting complex traffic analytic outputs and the reliance on computationally restrictive or expert-

driven methodologies that limit accessibility for general stakeholders (Dodge et al., 2021; Showribabu et al., 2025). The framework developed in this study responds to these challenges by integrating open traffic data, spatial statistical methods, and geovisual analytics within an explainable workflow. As such, it demonstrates that meaningful insights into congestion behavior can be derived without dependence on large-scale infrastructure or black box modeling approaches. Beyond methodological considerations, this research also extends existing work by providing a comparative analysis of spatial clustering across multiple metropolitan regions, offering a broader perspective on how congestion patterns vary by time and space. The use of interactive visualization further aligns with prior findings that emphasize the importance of exploratory visual analysis for understanding dynamic spatial phenomena, while promoting the role of geographic context as a unifying framework for integrating diverse traffic datasets (Luo et al., 2023; van Landesberger et al., 2016; Hart & Dolbear, 2007). These contributions advance the application of spatially grounded analytical techniques for transportation research and establish their use in real-world planning.

While the findings provide strong evidence of spatial structure in congestion patterns, several limitations should be considered when interpreting the results. The analysis relied on archived snapshot data collected over a two-week period, which captures representative peak conditions but does not fully account for longer term variability such as seasonal trends or special events. Additionally, while the clustering and spatial statistical methods provide valid outputs, they remain descriptive in nature and do not predict future congestion conditions. These limitations present clear opportunities for expansion. Extending the temporal scope of the dataset would improve how congestion patterns evolve over time. Using the underlying dataset to develop a predictive model could further increase the analytical value of the research by forecasting congestion hotspots which can be used for proactive traffic management strategies. From a systems perspective, transitioning the data pipeline to a fully cloud-based architecture would allow data ingestion, transformation, clustering, and visualization processes to be executed programmatically on an automated schedule, reducing reliance on manual data collection. Including such an approach would increase the operational viability of the framework for real-world deployment.

5 CONCLUSION

The goal of this research was to quantify and interpret the spatial structure of roadway congestion in California metropolitan regions. Because traffic conditions are shaped by roadway connectivity and travel demand, congestion would not be expected to behave as a purely random process in practice. Being so, the primary contributions of this research were to measure the degree to which congestion exhibited spatial dependence, identify where localized clustering occurred, and evaluate how these behaviors varied by location and time. Positive spatial autocorrelation observed in all study areas confirms that congested roadway segments are associated with nearby segments exhibiting similar conditions, although the intensity of this relationship fluctuated by time of day and urban area. Evening peak periods were generally associated with more concentrated and corridor-based congestion, while some morning and weekend influenced conditions showed weaker spatial structure. Local spatial statistical analysis revealed that congestion was organized not only around identifiable hotspot locations, but also around transitional zones where congestion changed between neighboring segments. These findings establish that congestion is inherently spatial in nature and formed by both travel behavior and the structure of urban transportation networks.

Beyond quantifying these spatial patterns, this research advances the field by confirming that spatial statistical methods and geovisual analytics can be combined within an accessible framework to promote interpretation of complex traffic data. Using open data sources, clustering techniques and interactive visualization creates a scalable approach for exploring congestion conditions within the CongestionView CA application without relying on intensive modeling systems. This research contributes new insight by

linking quantitative measures of spatial dependence with user friendly exploratory visualization, creating a solution designed to support both technical and non-technical audiences in examining where and how congestion occurs. Although formal user evaluation remains necessary to determine how effectively the interface improves comprehension and decision-making, the research successfully demonstrates that congestion structure can be measured, mapped, and interactively explored through a cumulative spatial data science workflow.

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